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Lecture 6: Probabilistic and First Moment Methods

Content.

1. Probabilistic Method
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1 Probabilistic Method

The idea is to show that objects satisfying a certain probability exists.

Theorem 1. Let $v_1, v_2, ..., v_n$ be unit vectors in $\mathbb{R}^n$ and there exists signs $\varepsilon_1, ..., \varepsilon_n \in \{-1, 1\}$ such that $||\varepsilon_1 v_1 + \varepsilon_2 v_2 ... + \varepsilon_1 v_n|| \leq \sqrt{n}$.[1]

Theorem 1, at first glance, looks like a difficult geometric problem. In what follows, we will provide an elegant probabilistic solution to it.

Theorem 2. Let X be a random variable with finite expectation. Then, for any μ, $\mathbb{E}X \leq \mu$ implies $\mathbb{P}[X \leq \mu] > 0$.

Proof of Theorem 2. Note that $\{X \leq \mu\} = \cap_{n=1}^{\infty} \{X < \mu + 1/n\}$. Suppose, for the sake of contradiction, that $\mathbb{E}X \leq \mu$ but $\mathbb{P}[X \leq \mu] = 0 = \lim_{n\to\infty} \mathbb{P}[X < \mu + 1/n]$. Thus, $\mathbb{P}[X < \mu + 1/n] < \epsilon$ for some n large. Notice that $X =$

[1] Given $v \in \mathbb{R}^n$, $\|v\|$ denotes is Euclidean (ℓ_2) norm, i.e., $\|v\| = \sqrt{\sum_i v_i^2}$.

$X\mathbb{I}_{X<\mu+1/n} + X\mathbb{I}_{X\geq\mu+1/n}$. Since $\mu \geq \mathbb{E}X$, we have

$$\begin{aligned}
\mu &\geq \mathbb{E}X\mathbb{I}_{X<\mu+1/n} + \mathbb{E}X\mathbb{I}_{X\geq\mu+1/n} \\
&\geq \mu \cdot \mathbb{E}[\mathbb{I}_{X<\mu+1/n}] + (\mu+1/n)\cdot\mathbb{E}[\mathbb{I}_{X\geq\mu+1/n}] \\
&= \mu\mathbb{P}[X<\mu+1/n] + (\mu+1/n)\mathbb{P}[X\geq\mu+1/n] \\
&= (\mu+1/n) - \frac{1}{n}\mathbb{P}[X<\mu+1/n] \\
&= \mu + \frac{1}{n}(1-\epsilon) \\
&> \mu
\end{aligned}$$

since $P[X \geq \mu+1/n] = 1 - P[X<\mu+1/n]$ and $P[X<\mu+1/n] < \epsilon$.

Proof of Theorem 1. Let us introduce randomness. Suppose that $\varepsilon_i \overset{\text{iid}}{\sim} \text{Rad}(1/2)$, i.e. $\mathbb{P}[\varepsilon_i = 1] = \mathbb{P}[\varepsilon_i = -1] = 1/2$. Let $X = X(\varepsilon_1, ..., \varepsilon_n) = ||\varepsilon_1 v_1 + ... + \varepsilon_n v_n||_2^2$. Note that

$$X = \sum_{i=1}^{n} \varepsilon_i^2 ||v_i||_2^2 + 2\sum_{i<j} \varepsilon_i\varepsilon_j v_i^T v_j$$

and that

$$\begin{aligned}
\mathbb{E}X &= \sum_{i=1}^{n} \mathbb{E}[\varepsilon_i^2] + 2\sum_{i<j}\mathbb{E}[\epsilon_i\epsilon_j]v_i^T v_j \\
&= n
\end{aligned}$$

So there is a realization $\varepsilon_1, ..., \varepsilon_n$ for which $||\sum_{i=1}^n \varepsilon_i v_i|| \leq \sqrt{n}$.

Observation. $\sqrt{n}$ is in general tight. If $v_i = e_i$, i-th element of the standard basis, then $||\sum_i \varepsilon_i e_i|| = ||(\epsilon_1, ..., \epsilon_n)|| = \sqrt{n}$.

1.1 Probabilistic Method for Random Graphs

Definition 1. Given a graph $G = (V, E)$, a clique is a subset $S \subseteq V$ of vertices for which all edges among vertices of S are present.

Definition 2. Suppose all edges are colored with one of q colors. A monochromatic clique is a clique whose edges have the same color.

Theorem 3. *(Erdős)* Let K_n be the complete graph on n vertices. It has all $\binom{n}{2}$ edges present. Suppose $k \geq 3$ and $n \leq 2^{k/2}$. Then, it is possible to color the edges with two colors such that there is no monochromatic clique of size k.

Proof of Theorem 3. We color each edge randomly and independently with probability $1/2$. Let us index cliques of size k, $e_1, e_2, ..., e_{\binom{n}{k}}$. Let $E_i = \{e_i \text{ is monochromatic}\}$. Note that $\mathbb{P}[E_i] = 2^{-\binom{k}{2}} \cdot 2$. Let $\varepsilon = \cap_{i=1}^{\binom{n}{k}} E_i^c$. We want to show that $\mathbb{P}[\varepsilon] > 0 \iff \mathbb{P}[\varepsilon^c] < 1$. Note that $\varepsilon^c = \cup_{i=1}^{\binom{n}{k}} \mathbb{P}[E_i]$. We will use the fact that $\binom{n}{k} \leq \left(\frac{en}{k}\right)^k$. Notice that

$$\begin{aligned}
\mathbb{P}[\varepsilon^c] &\leq \binom{n}{k} \mathbb{P}[E_i] \\
&\leq \left(\frac{en}{k}\right)^k \cdot 2 \cdot 2^{-k(k+1)/k} \\
&\leq \left(\frac{e}{k}\right)^k \cdot 2^{k^2/2} \cdot 2 \cdot 2^{-k^2/2} \cdot 2^{k/2} \\
&= 2 \cdot \left(\frac{e\sqrt{2}}{k}\right)^k
\end{aligned}$$

For $k > 5$, this is less than 5. For $k = 3, 4$, we can check by hand.

2 First Moment Method

The first moment method is a very powerful tool to prove the "absence" of certain random structures.

Markov's Inequality: Let X be a random variable that is almost surely non-negative. Then, for all $a > 0$, $\mathbb{P}[X \geq a] \leq \frac{1}{a}\mathbb{E}[X]$.

Proof. Note that

$$\begin{aligned}
X &= X \cdot \mathbb{I}_{X \geq a} + X \cdot \mathbb{I}_{X < a} \\
&\geq X \cdot \mathbb{I}_{X \geq a} \\
&\geq a \cdot \mathbb{I}_{X \geq a}
\end{aligned}$$

□

Let N be the random variable counting the number of structures with a given property. We want $\mathbb{P}[N = 0] \to 1$. Equivalently, $\mathbb{P}[N \neq 0] \to 0$. By Markov's inequality, $\mathbb{P}[N \neq 0] = \mathbb{P}[N \geq 1] \leq \mathbb{E}[N]$. So if $\mathbb{E}[N] \to 0$, then we finish the conclusion.

Theorem 4. Suppose m balls are thrown uniformly at random to n bins. Then, $\mathbb{P}[\text{there exists an empty bin}] \to 0$ as $n \to \infty$, provided $m > (1 + \epsilon)n \log n$.

Proof: Let $N = \sum_{i=1}^{n} I_i$ be the number of empty bins. Here, $I_i = \begin{cases} 1, & \text{if the i-th bin is empty} \\ 0, & \text{otherwise} \end{cases}$.

Note that

$$\begin{aligned} \mathbb{E}[I_i] &= \left(1 - \frac{1}{n}\right)^m \\ &= \mathbb{P}[I_i = 1] \end{aligned}$$

and that

$$\begin{aligned} \mathbb{E}[N] &= \sum_{i=1}^{n} \mathbb{E}[I_i] \\ &= n\left(1 - \frac{1}{n}\right)^m \\ &= \exp\left(\log(n) + m\log\left(1 - \frac{1}{n}\right)\right) \\ &= \exp\left(\log(n) + (1+\epsilon)n\log(n)\left(-\frac{1}{n}\right)(1 + o_n(1))\right) \\ &= \exp\left(-\epsilon\log(n)(1 + o(1))\right) \\ &= n^{-\epsilon(1+o(1))} \to 0 \end{aligned}$$

A more interesting application of the first moment method is to an important model in theoretical computer science.

Random k-SAT: A constraint satisfaction problem.
Model:

- $x_1, x_2, ..., x_n \in \{0, 1\}$. These are boolean variables.
- For fixed integer $k \in \mathbb{Z}^+$: k-clause.
- Choose $i_1, i_2, ..., i_k$ uniformly at random from $\{1, ..., n\}$. Let y_{i_j} be equal to x_{i_j} with probability $\frac{1}{2}$ and equal to x'_{i_j}, also with probability $\frac{1}{2}$. (Namely, negate each x_{i_j} with probability $\frac{1}{2}$). Consider the k-clause

$$e = y_{i_1} \vee y_{i_2} \cdots \vee y_{i_k}.$$

- Sample $M = \alpha n$ k-clauses $e_1, ..., e_M$ and independently.
- Construct the formula $\Phi : e_1 \wedge e_2 \wedge ... \wedge e_M$.

Question: Fix k. As $n \to \infty$, when do satisfying assignments exist? i.e. $(x_1, ..., x_n) \in \{0,1\}^n$ for which $\Phi(x_1, ..., x_n) = 1$.

Theorem 5 (Franco-Paull'83)**.** If $\alpha > 2^k \ln 2$, then with high probability there are no satisfying assignments.

Hint: Let $N = \sum_{x \in \{0,1\}^n} I_x$, where $I_x = \begin{cases} 1, \text{ if } x \text{ is satisfying} \\ 0, \text{ o.w.} \end{cases}$. Study $\mathbb{E}[N]$.